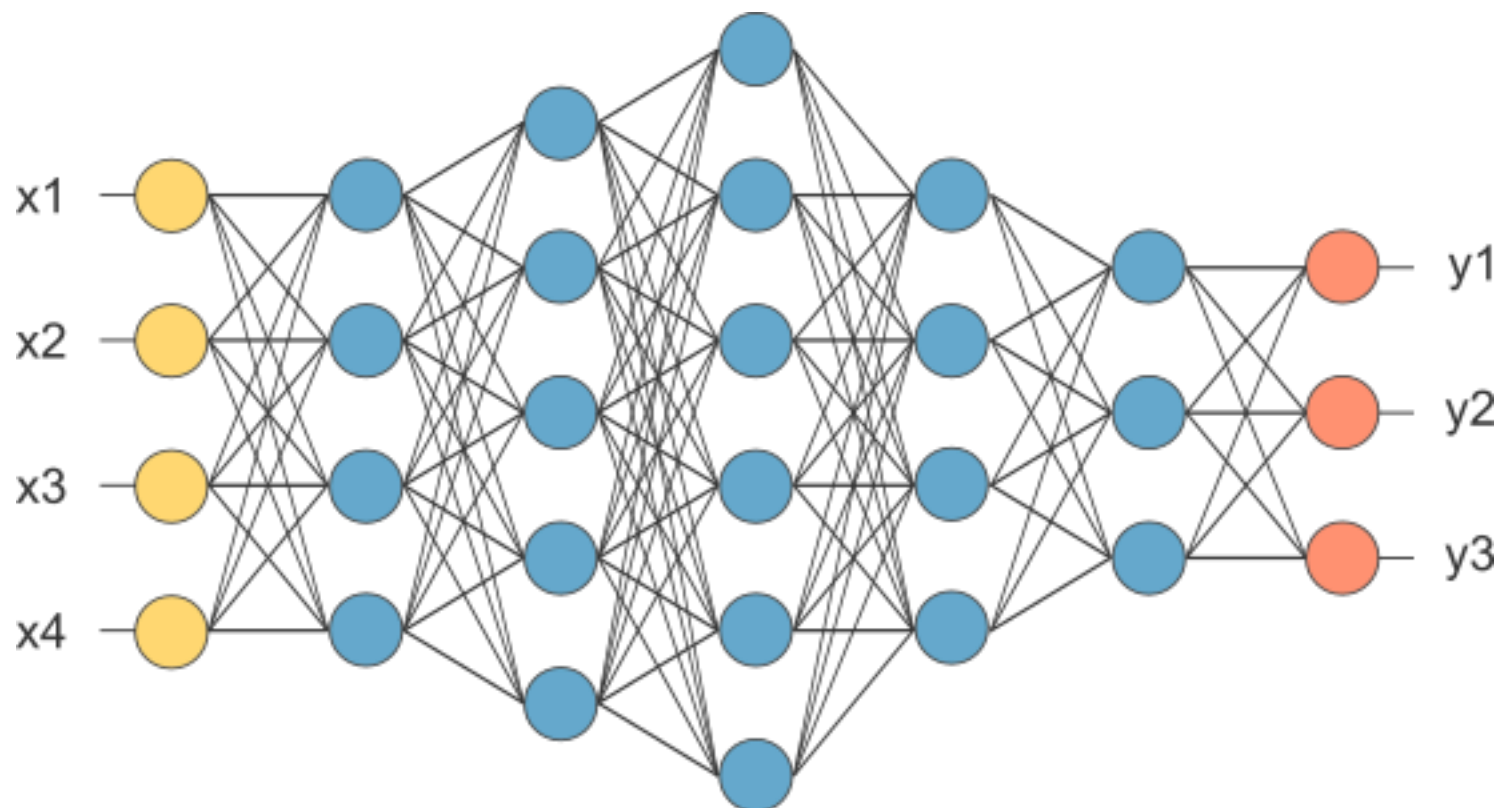


RNN, LSTM and GRU

CSE 4237

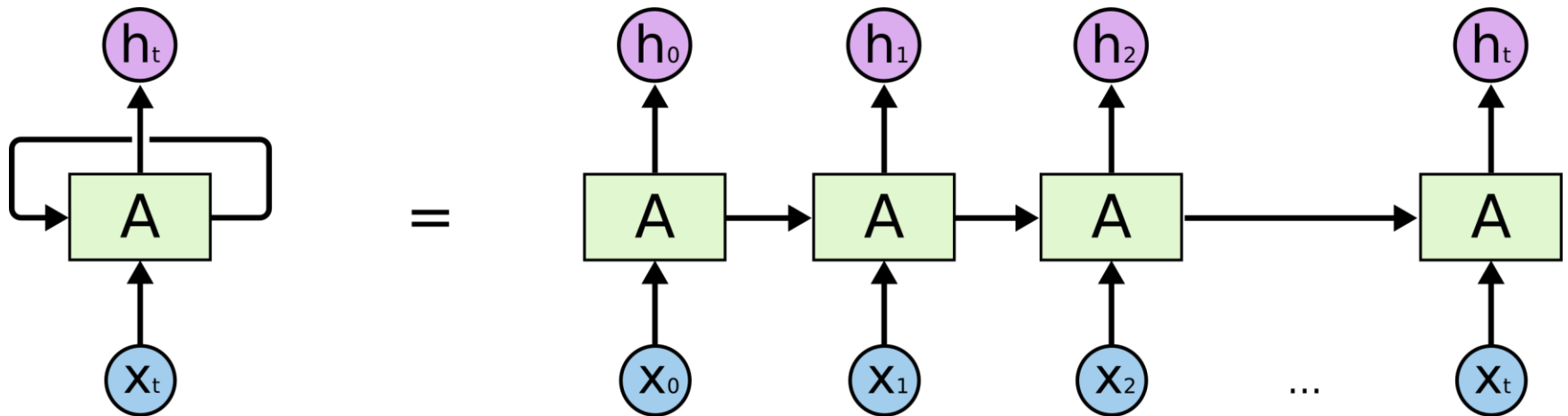
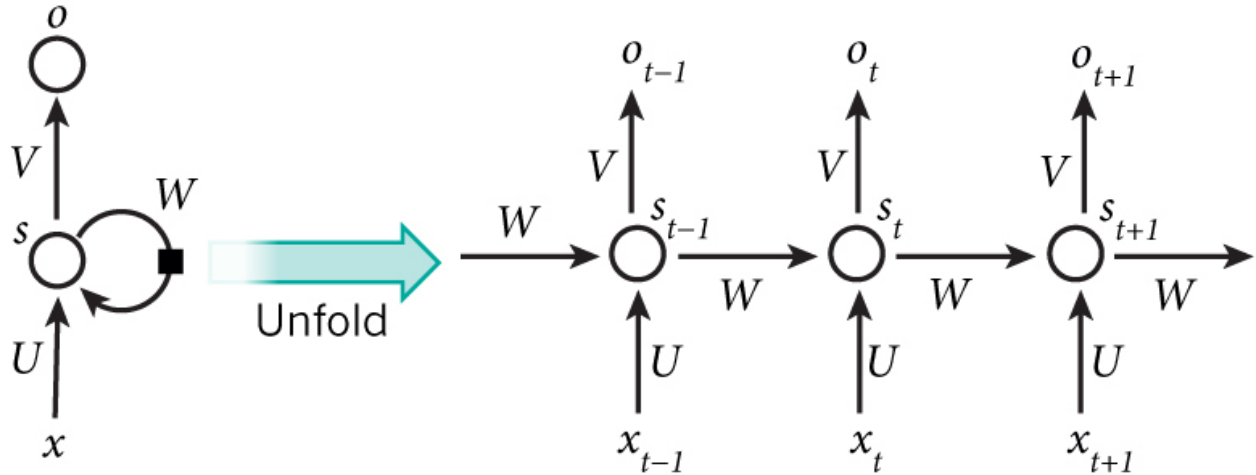
Soft Computing

Smaller Network: RNN



This is our fully connected network. If $x_1 \dots x_n$, n is very large and growing, this network would become too large. We now will input **one x_i at a time**, and **re-use the same edge weights**.

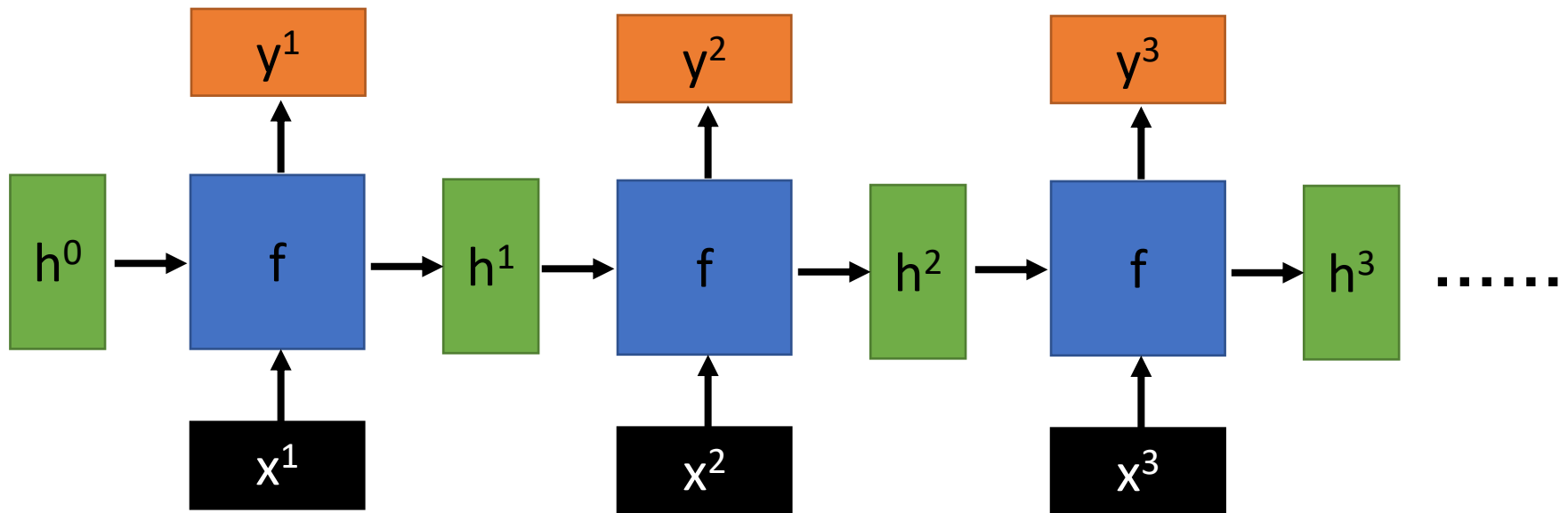
Recurrent Neural Network



How does RNN reduce complexity?

- Given function $f: h', y = f(h, x)$

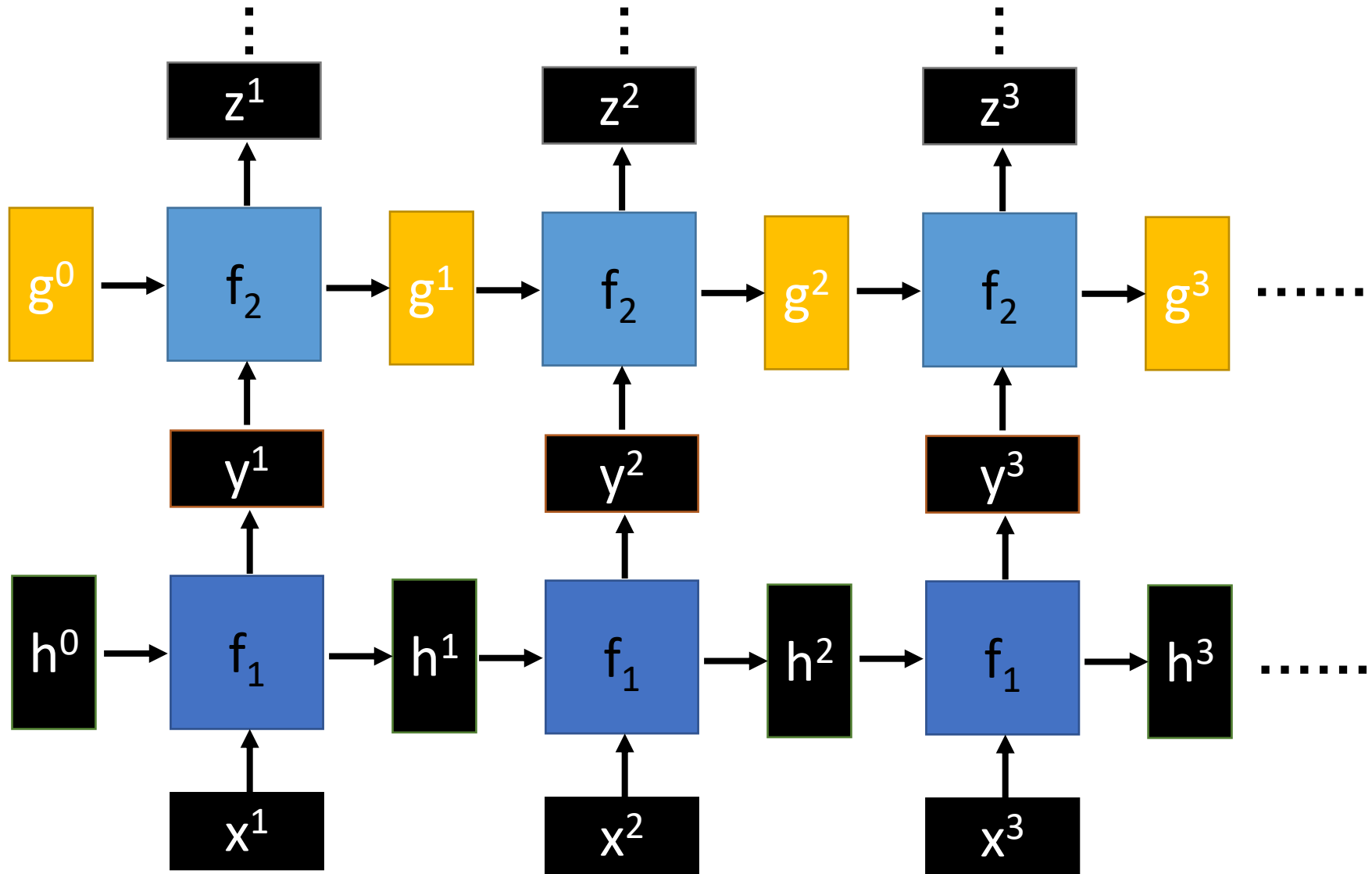
h and h' are vectors with the same dimension



No matter how long the input/output sequence is, we only need one function f . If f 's are different, then it becomes a feedforward NN. This may be treated as another compression from fully connected network.

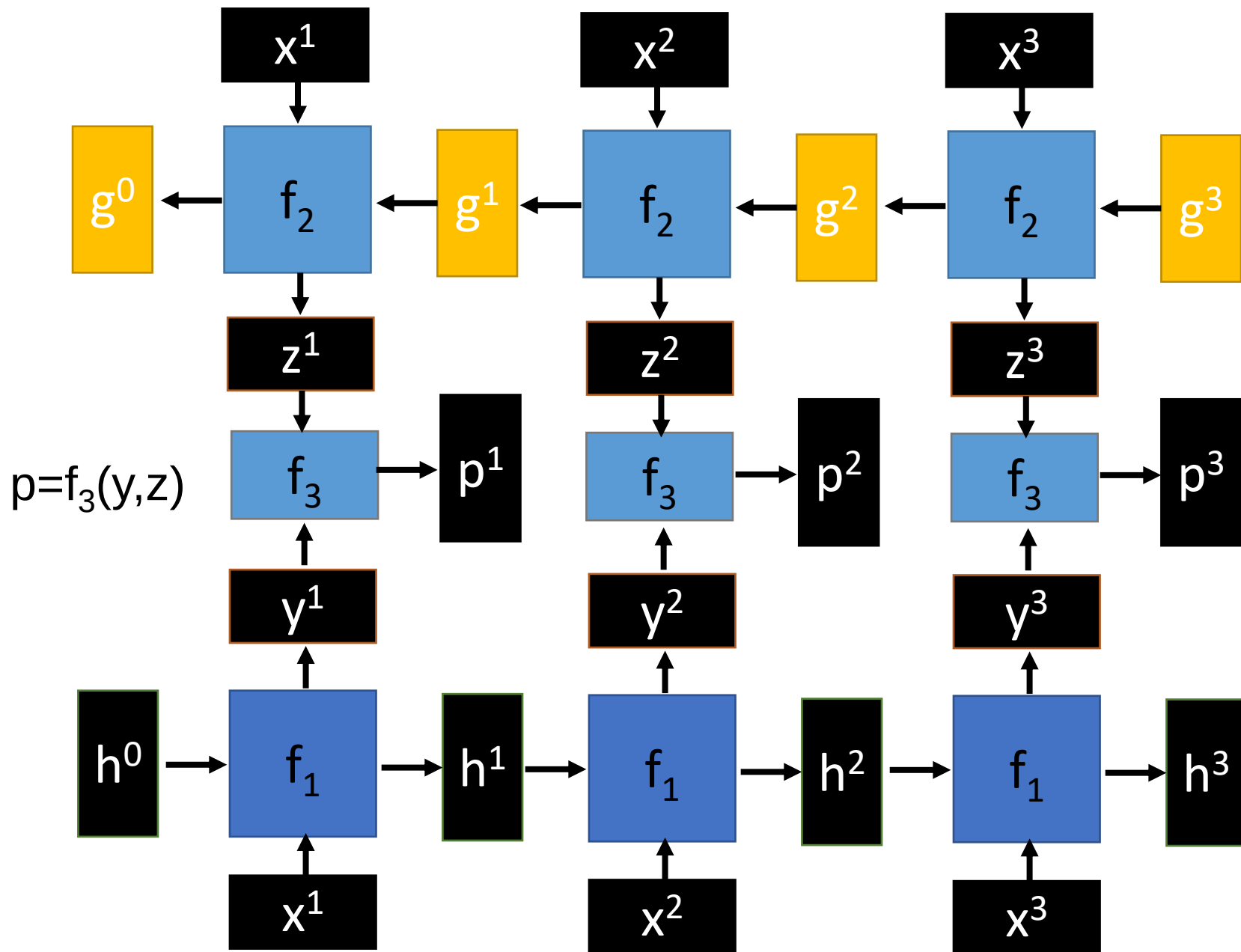
Deep RNN

$$h', y = f_1(h, x), \quad g', z = f_2(g, y) \quad \dots$$



Bidirectional RNN

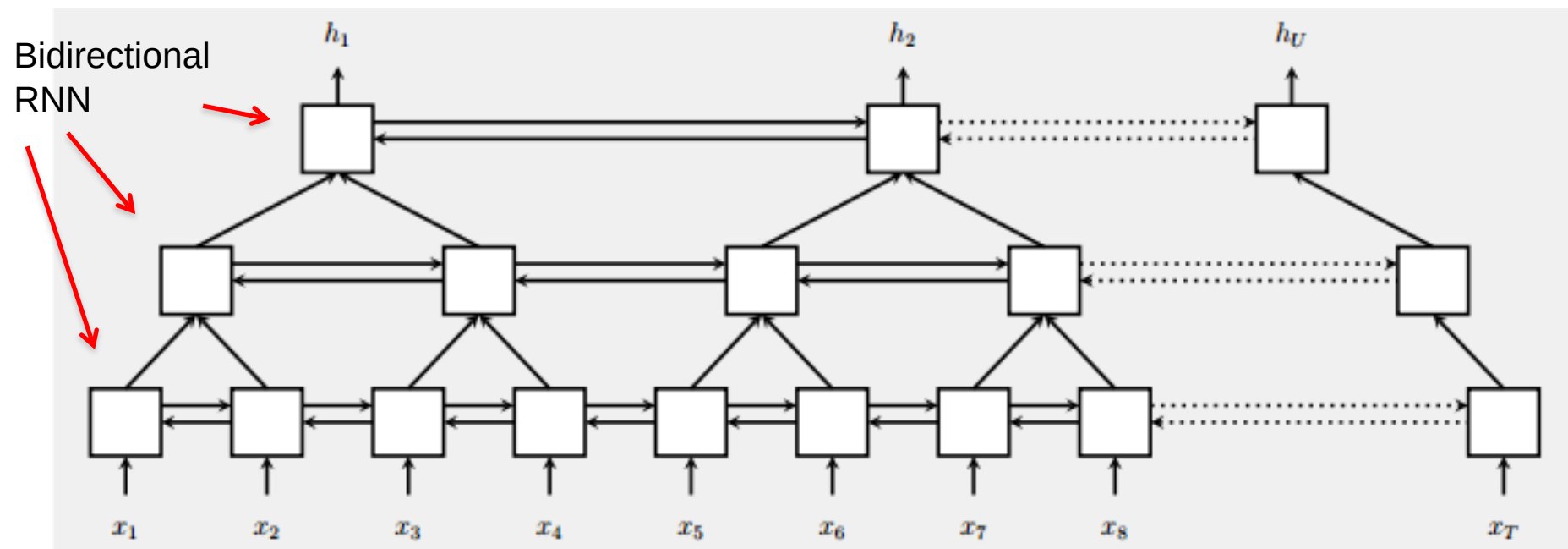
$$y, h = f_1(x, h) \quad z, g = f_2(g, x)$$



Pyramid RNN

Significantly speed up training

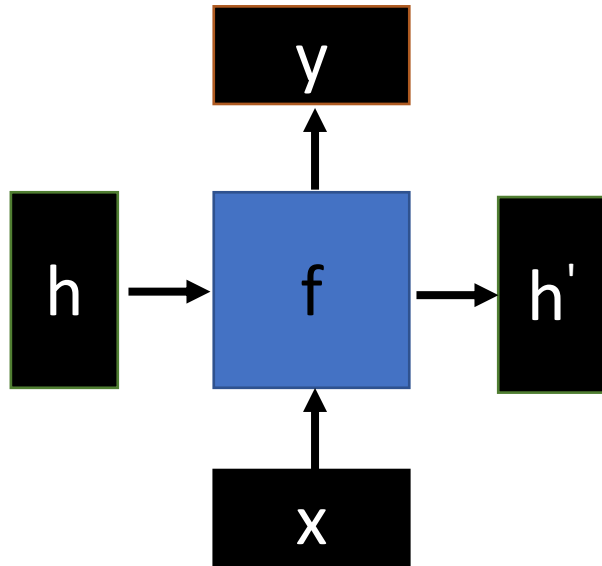
- Reducing the number of time steps



W. Chan, N. Jaitly, Q. Le and O. Vinyals, "Listen, attend and spell: A neural network for large vocabulary conversational speech recognition," ICASSP, 2016

Naïve RNN

- Given function $f: h', y = f(h, x)$



$$h' = \sigma(W^h h + W^i x)$$

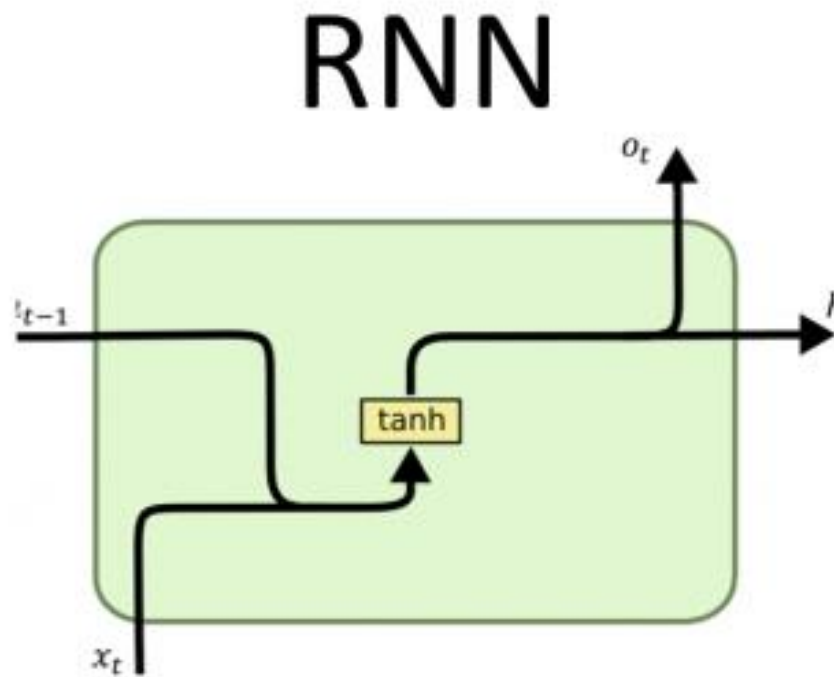
$$y = \sigma(W^o h')$$

softmax

Note, y is computed from h'

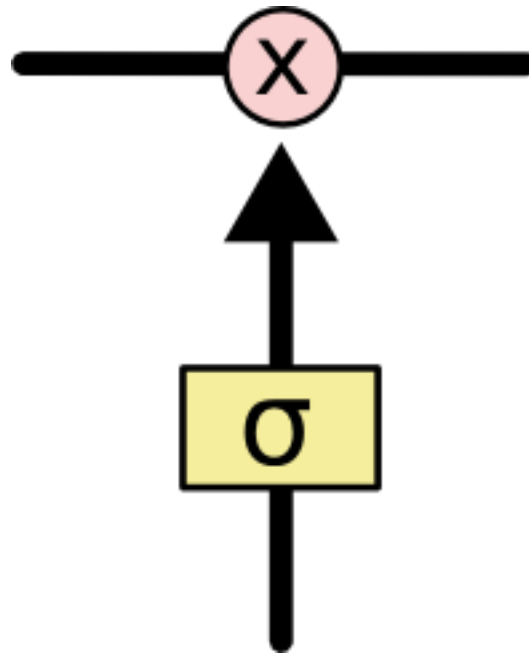
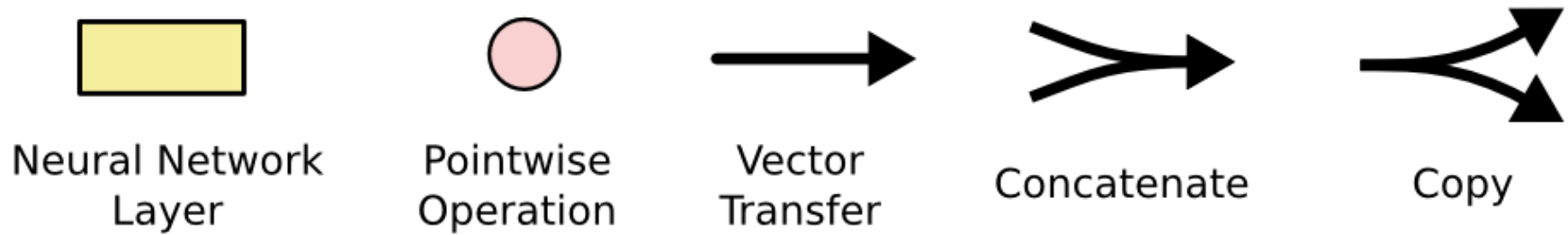
We have ignored the bias

Naïve RNN



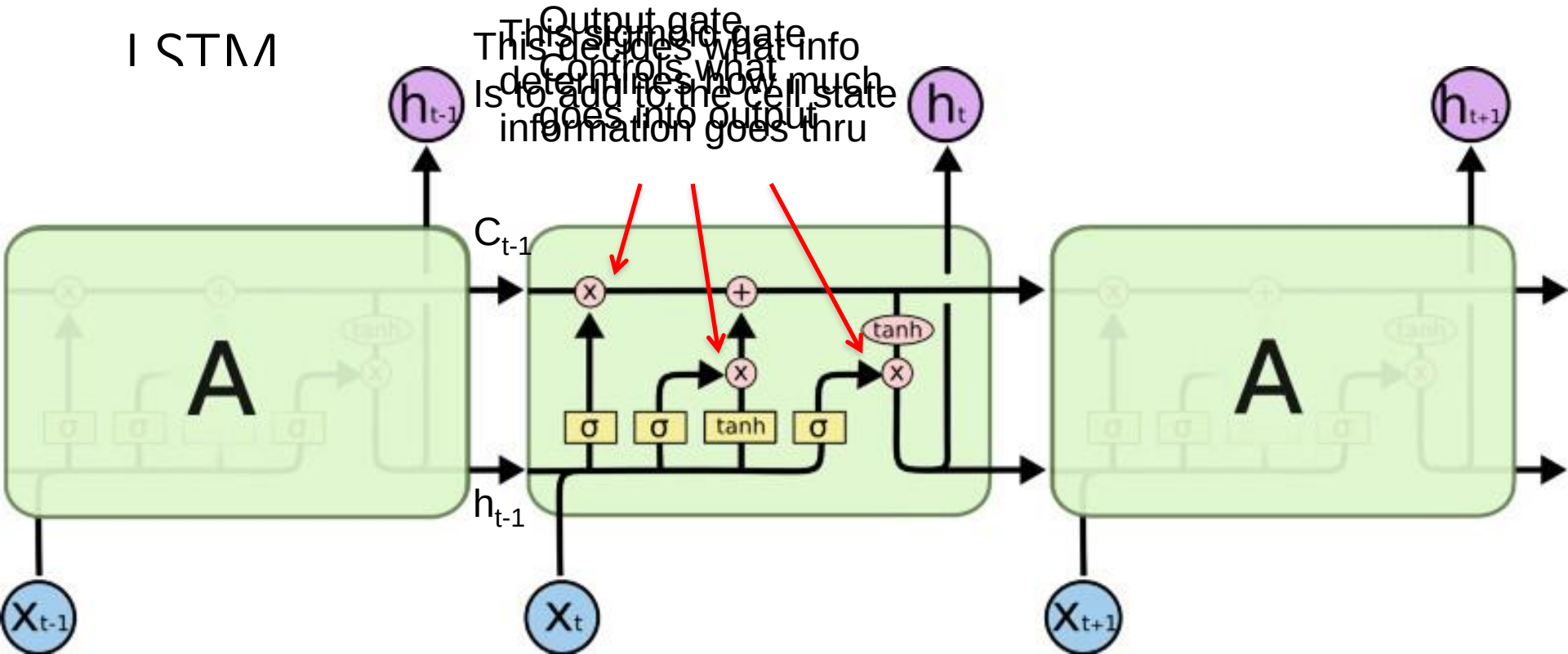
Problems with naive RNN

- When dealing with a time series, it tends to forget old information. When there is a distant relationship of unknown length, we wish to have a “memory” to it.
- Vanishing gradient problem.



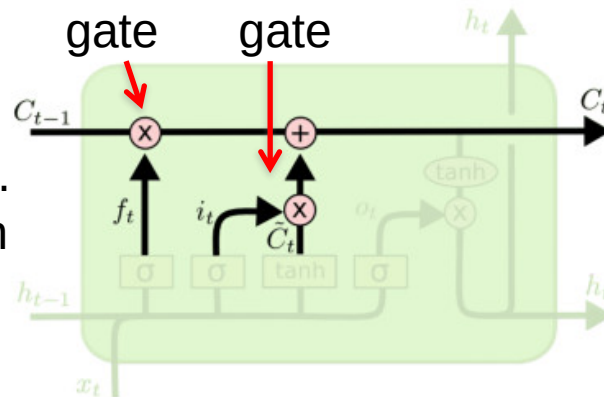
The sigmoid layer outputs numbers between 0-1 determine how much each component should be let through. Pink X gate is point-wise multiplication.

LSTM



Output gate
This decides what info
goes into the cell state
information goes thru

Forget gate
input gate



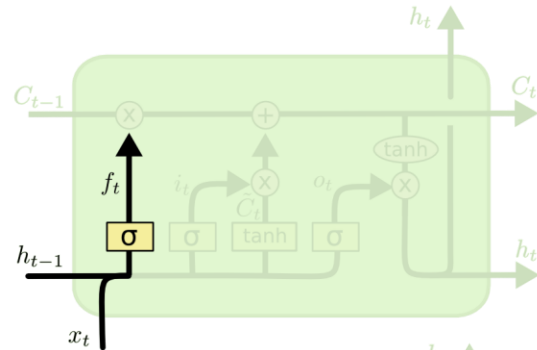
Why sigmoid or tanh:
Sigmoid: 0,1 gating as switch.
Vanishing gradient problem in
LSTM is handled already.
ReLU replaces tanh ok?

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

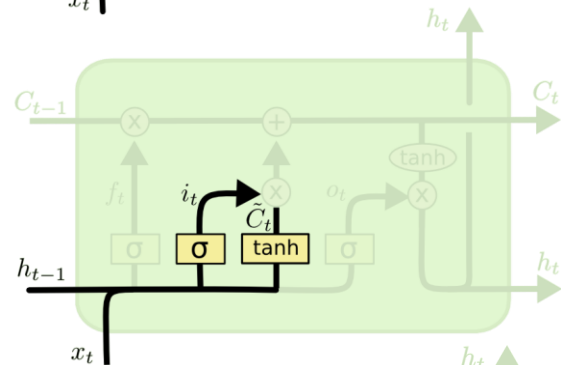
The core idea is this cell
state C_t it is changed
slowly with only minor
vanishing gradient problem in
LSTM is handled already.
easy for information to flow
ReLU replaces tanh ok?
along it unchanged.

LSTM

- Sigmoid specifically, is used as the gating function for the three gates (in, out, and forget) in LSTM, since it outputs a value between 0 and 1, and it can either let no flow or complete flow of information throughout the gates.
- On the other hand, **to overcome the vanishing gradient problem**, we need a function whose second derivative can sustain for a long range before going to zero. **Tanh is a good function** with the above property.
- A good neuron unit should be bounded, easily differentiable, monotonic (good for **convex optimization**) and easy to handle. If you consider these qualities, then you can use ReLU in place of the tanh function since they are very good alternatives of each other.
- But before making a choice for activation functions, you must know what the advantages and disadvantages of your choice over others are.

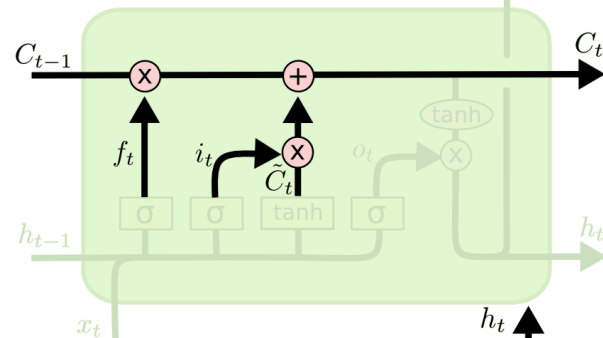


$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

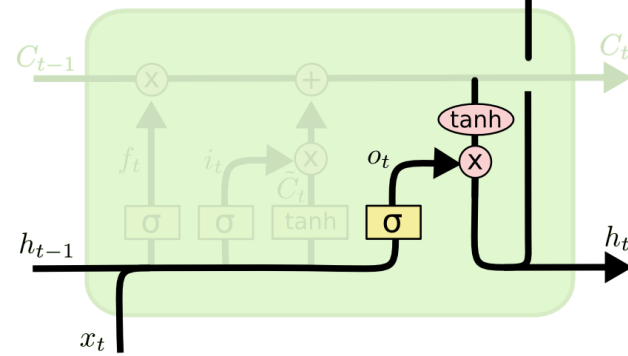


$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$



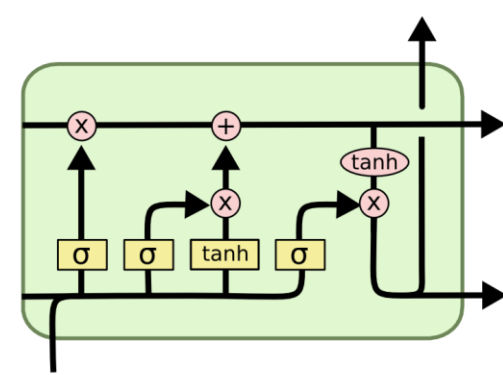
$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$



$$o_t = \sigma(W_o [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t * \tanh(C_t)$$

Decide what part of the cell state to output

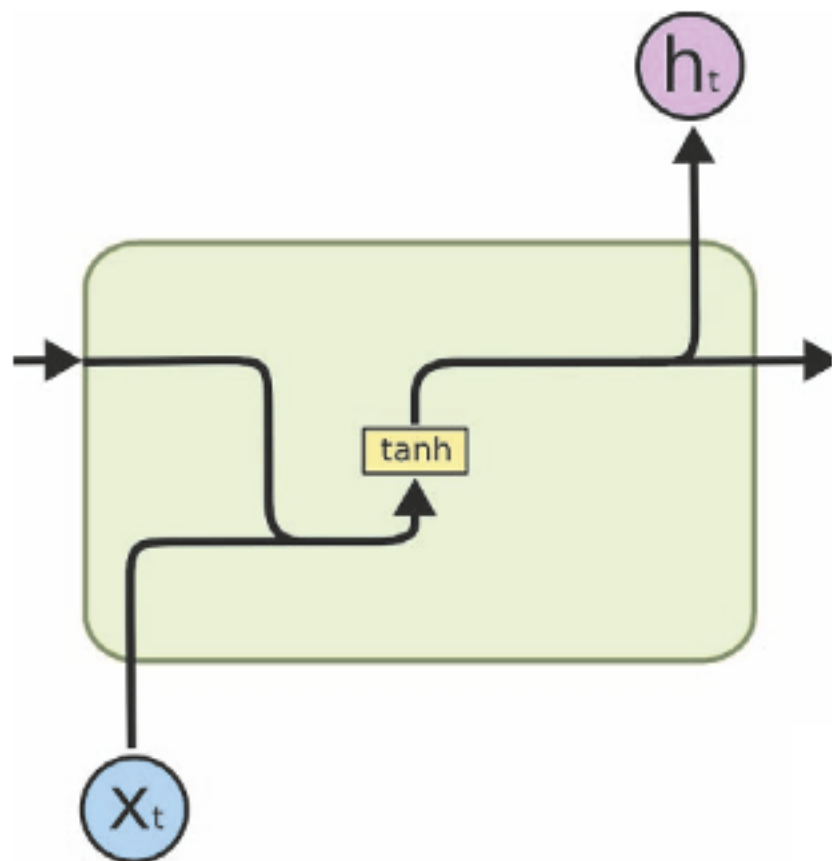


i_t decides what component is to be updated.

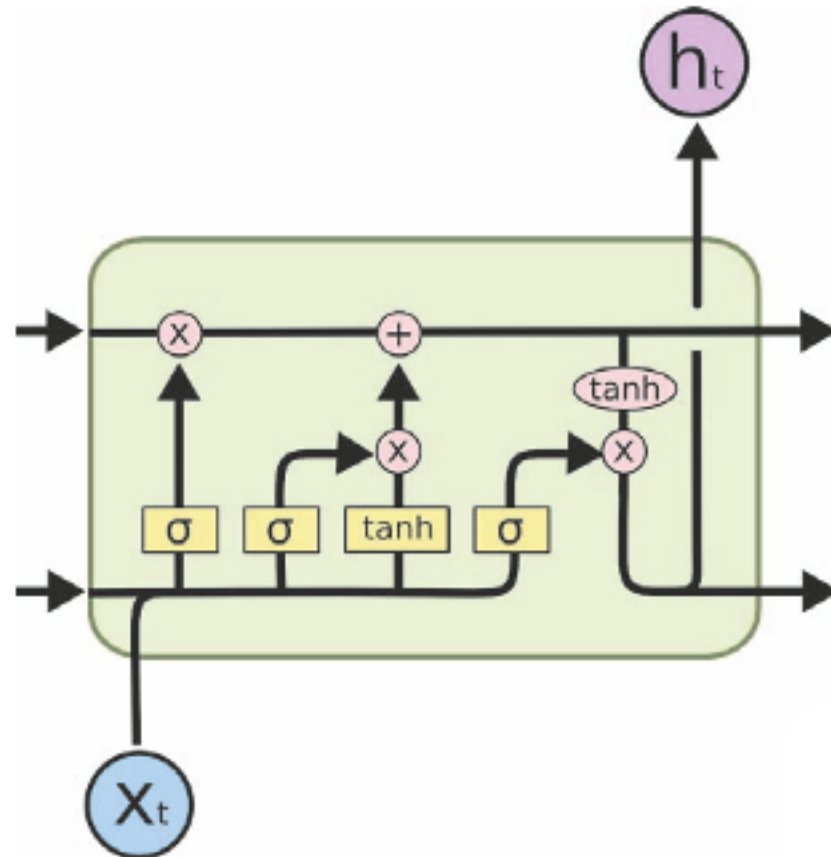
C'_t provides **change contents**

Updating the cell state

RNN vs LSTM

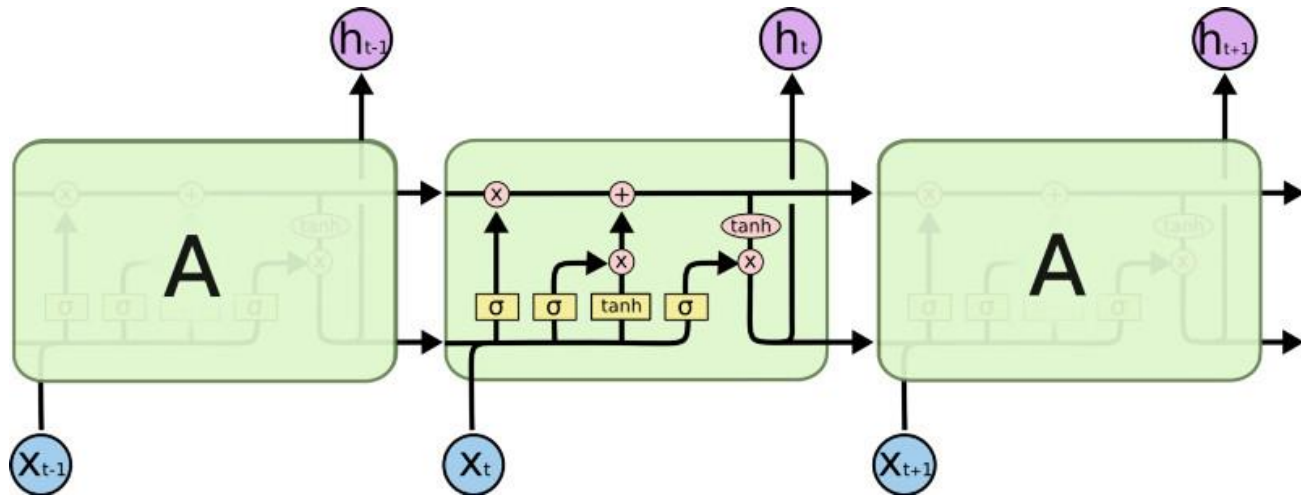


(a) RNN

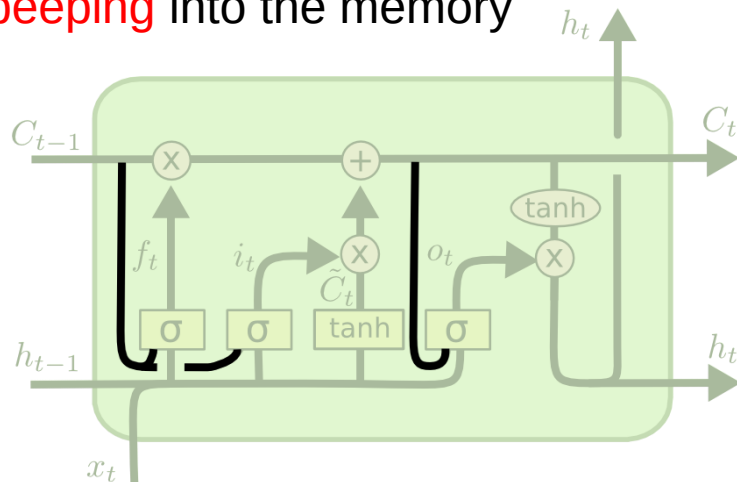


(b) LSTM

Peephole LSTM



Allows “**peeping** into the memory”

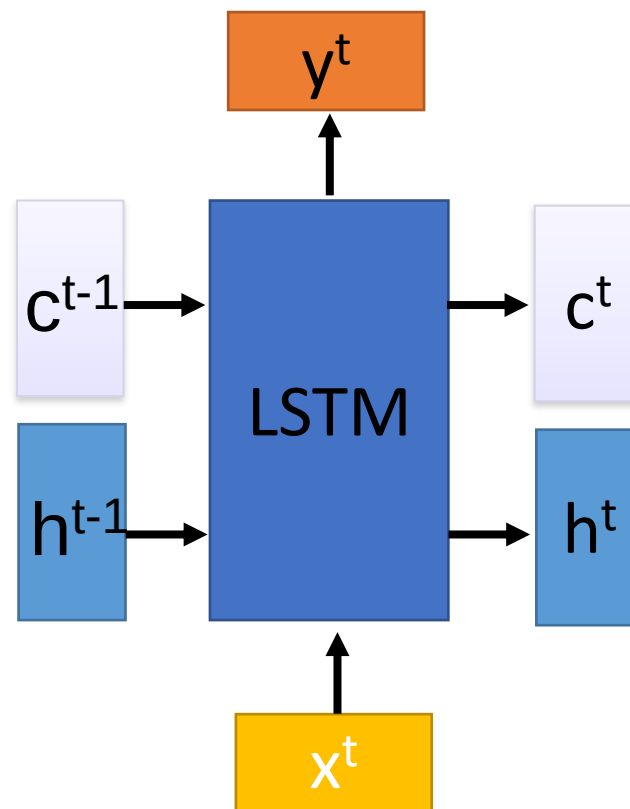
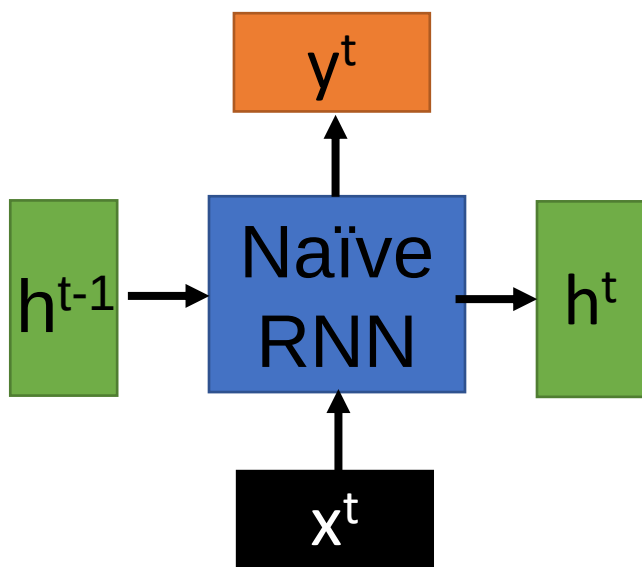


$$f_t = \sigma(W_f \cdot [C_{t-1}, h_{t-1}, x_t] + b_f)$$

$$i_t = \sigma(W_i \cdot [C_{t-1}, h_{t-1}, x_t] + b_i)$$

$$o_t = \sigma(W_o \cdot [C_t, h_{t-1}, x_t] + b_o)$$

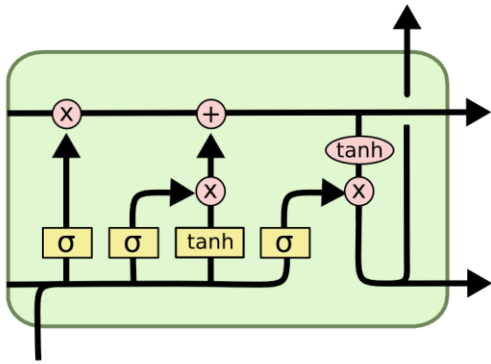
Naïve RNN vs LSTM



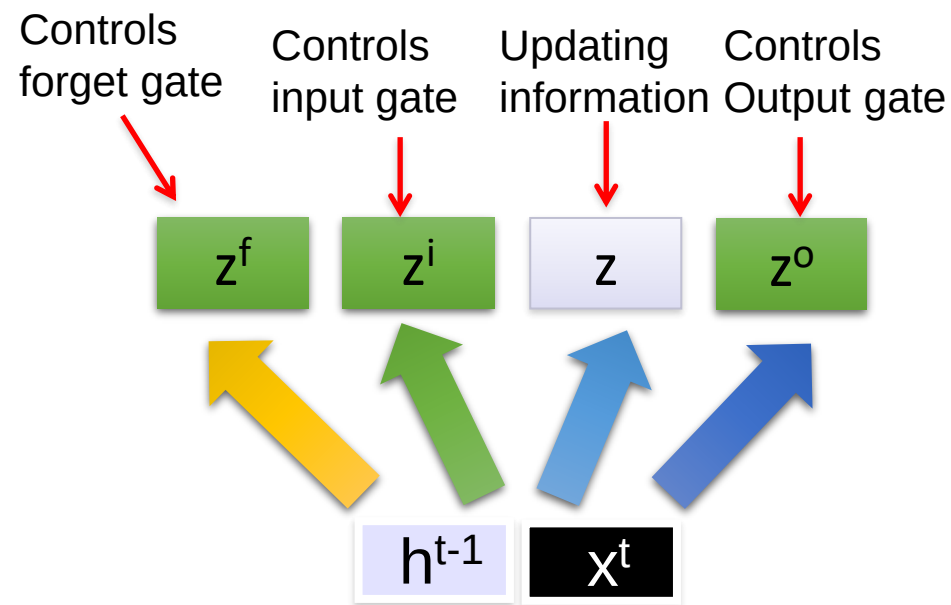
c changes slowly $\Rightarrow c^t$ is c^{t-1} added by something

h changes faster $\Rightarrow h^t$ and h^{t-1} can be very different

These 4 matrix computation should be done concurrently.



c^{t-1}



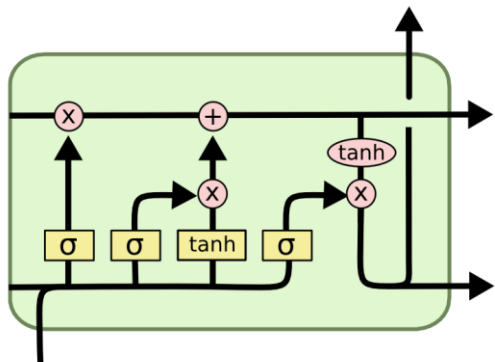
$$z = \tanh(W \begin{bmatrix} x^t \\ h^{t-1} \end{bmatrix})$$

$$z^i = \sigma(W^i \begin{bmatrix} x^t \\ h^{t-1} \end{bmatrix})$$

$$z^f = \sigma(W^f \begin{bmatrix} x^t \\ h^{t-1} \end{bmatrix})$$

$$z^o = \sigma(W^o \begin{bmatrix} x^t \\ h^{t-1} \end{bmatrix})$$

Information flow of LSTM

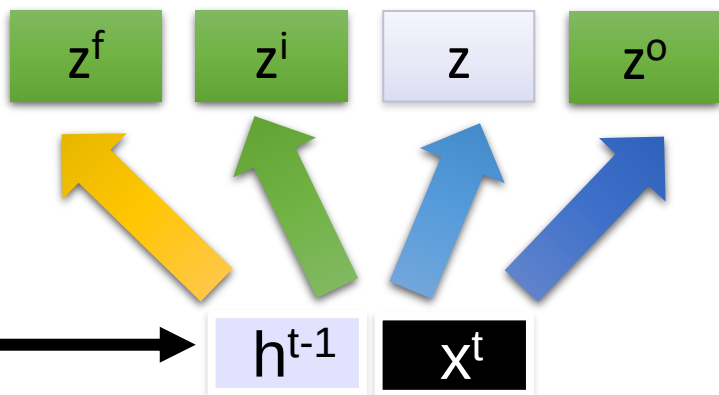


$$z = \tanh\left(W \begin{bmatrix} x^t \\ h^{t-1} \\ c^{t-1} \end{bmatrix} \right)$$

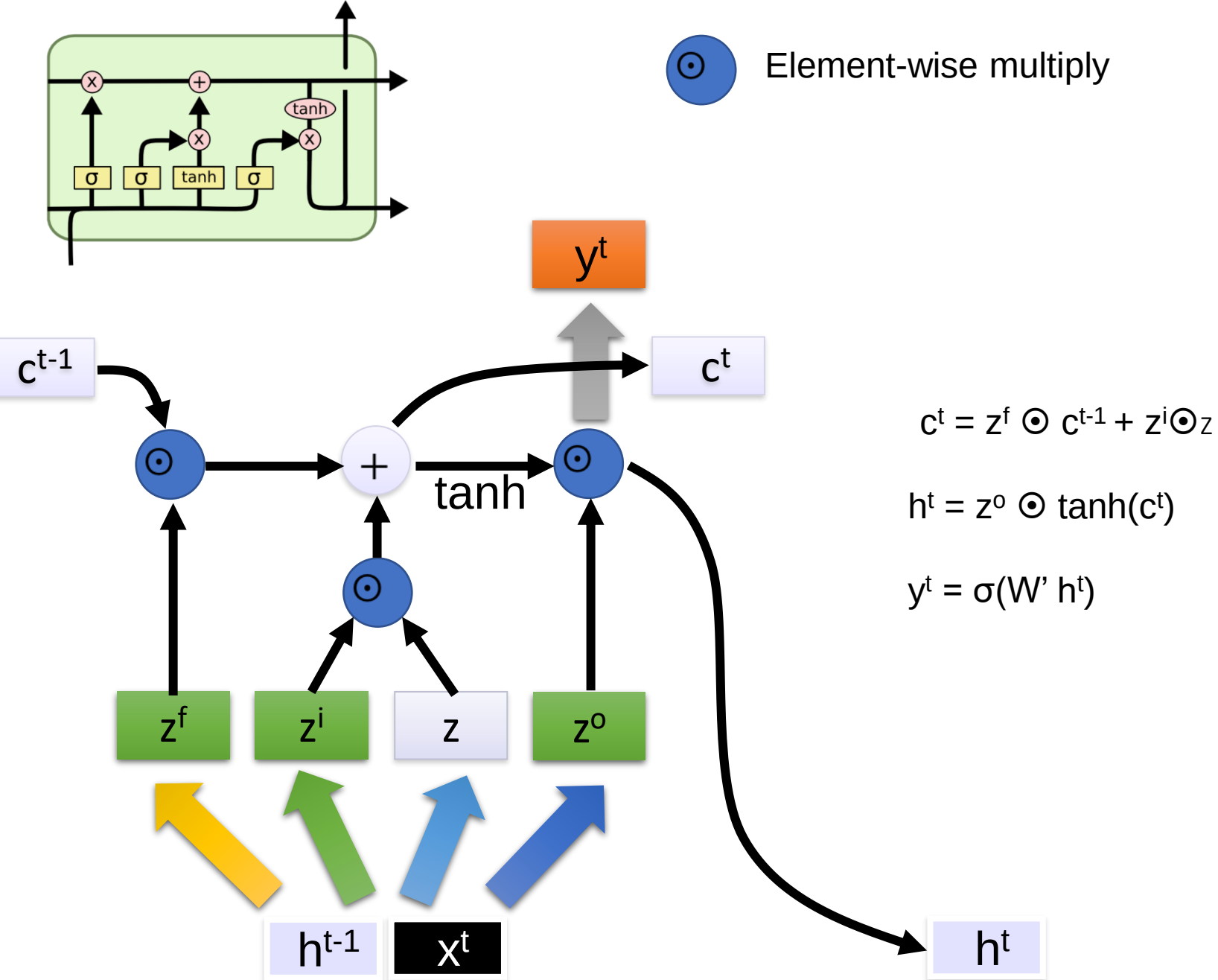
diagonal

z^o z^f z^i obtained by the same way

“peephole”

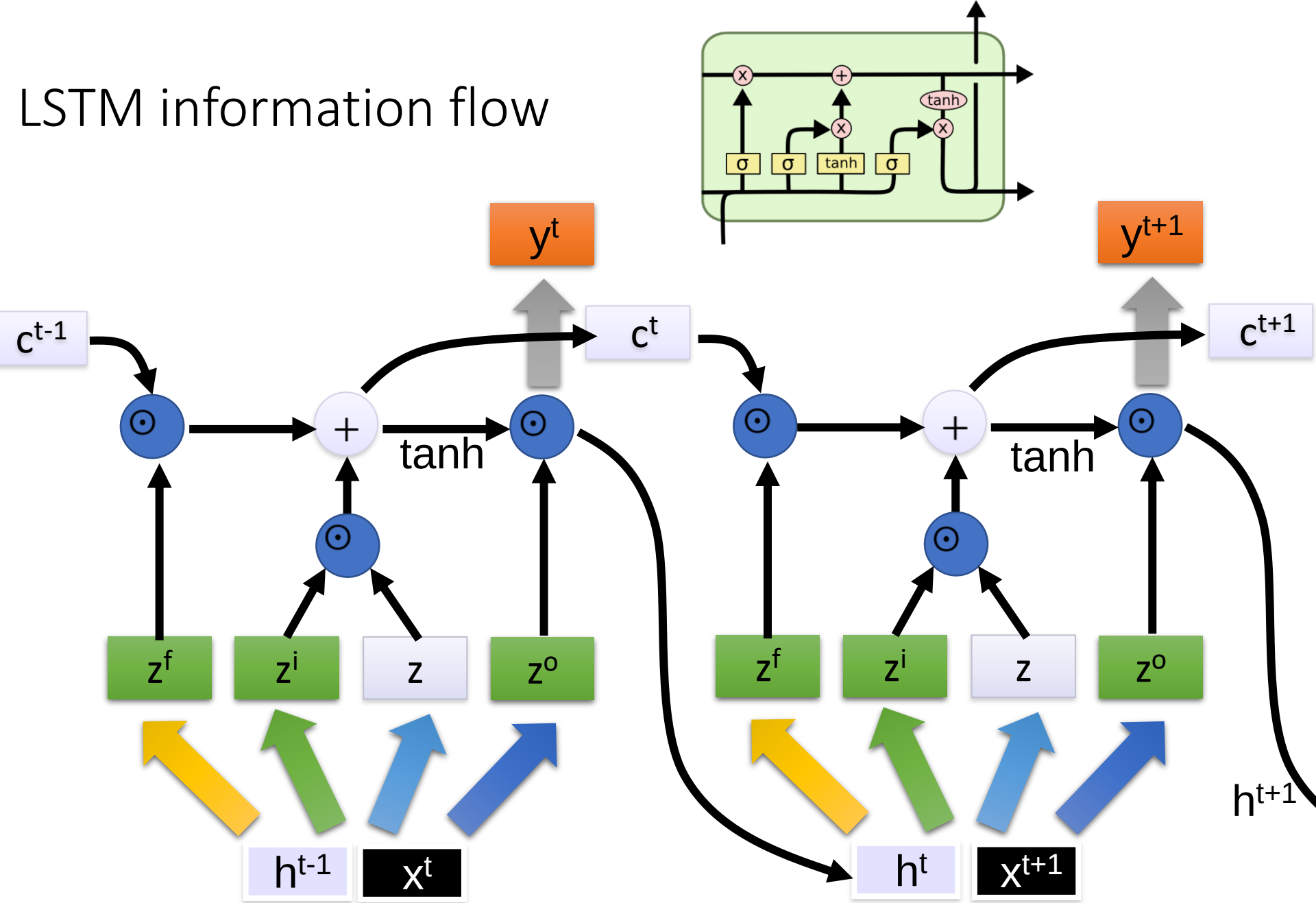


Information flow of LSTM



Information flow of LSTM

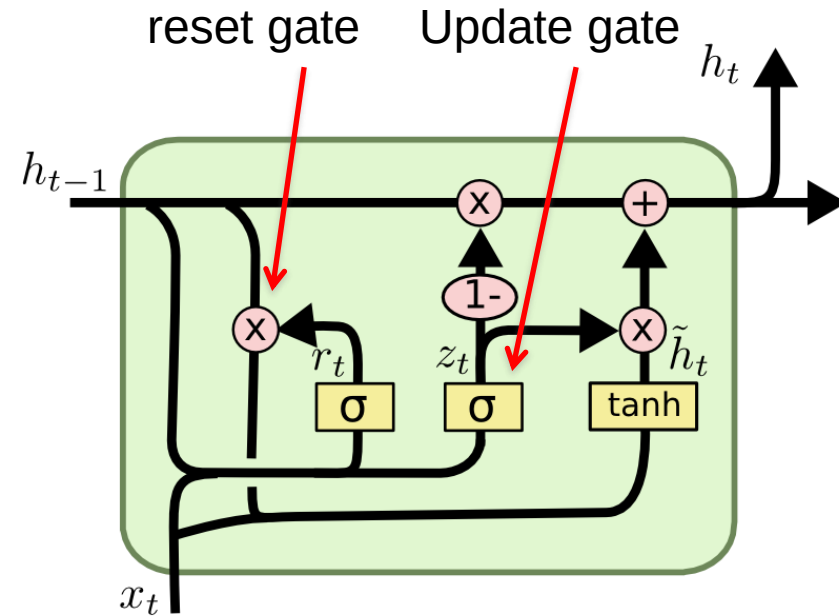
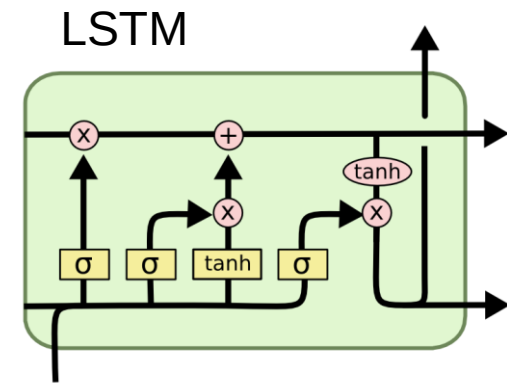
LSTM information flow



Information flow of LSTM

GRU – gated recurrent unit

(more compression)



$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t])$$

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t])$$

$$\tilde{h}_t = \tanh(W \cdot [r_t * h_{t-1}, x_t])$$

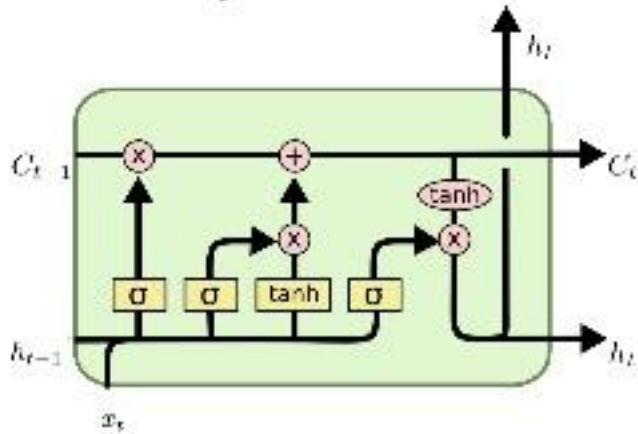
$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t$$

It combines the **forget** and **input** into a single **update gate**.
It also merges the cell state and hidden state. This is simpler than LSTM. There are many other variants too.

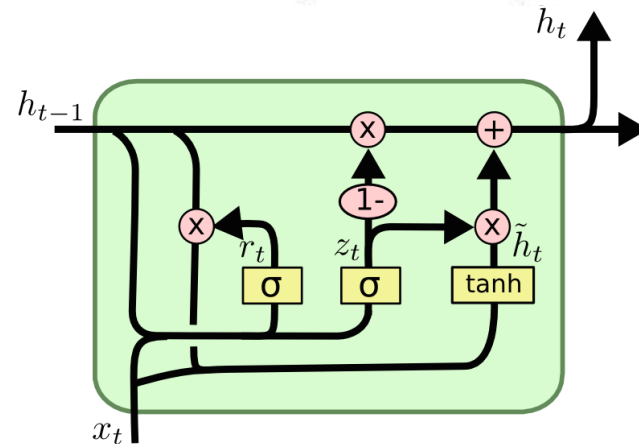
X,*: element-wise multiply

LSTM and GRU

- LSTM [Hochreiter&Schmidhuber97]



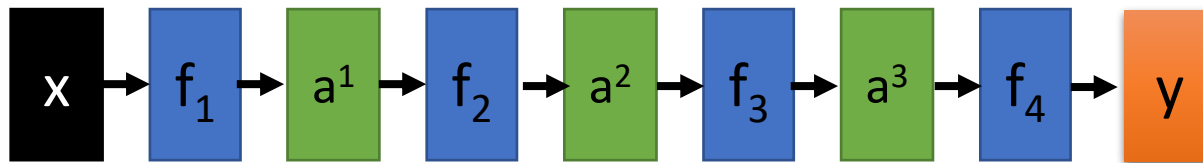
- GRU [Cho+14]



GRUs also take x_t and h_{t-1} as inputs. They perform some calculations and then pass along h_t . What makes them different from LSTMs is that **GRUs don't need the cell layer to pass values along**. The calculations within each iteration ensure that the h_t values being passed along **either retain a high amount of old information or are jump-started with a high amount of new information**.

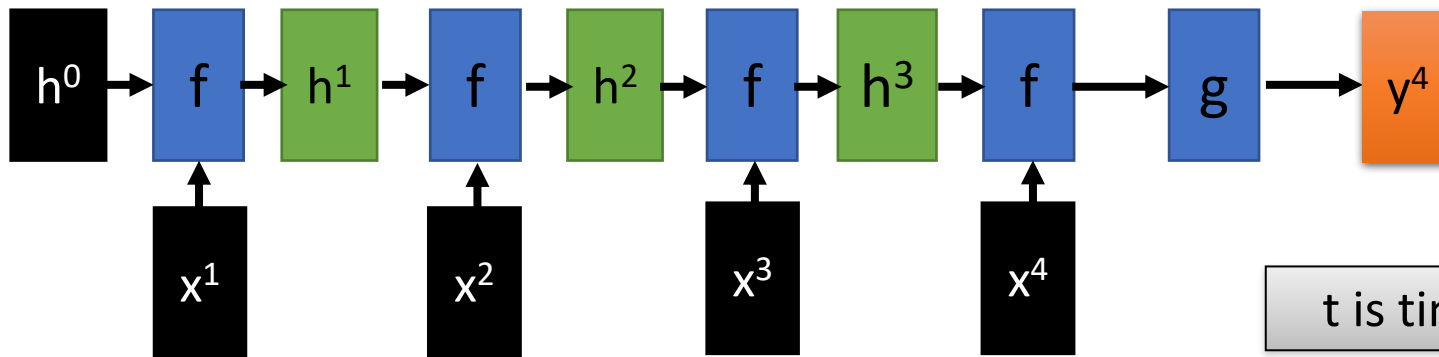
Feed-forward vs Recurrent Network

1. Feedforward network does not have input at each step
2. Feedforward network has different parameters for each layer



$$a^t = f_t(a^{t-1}) = \sigma(W^t a^{t-1} + b^t)$$

t is layer



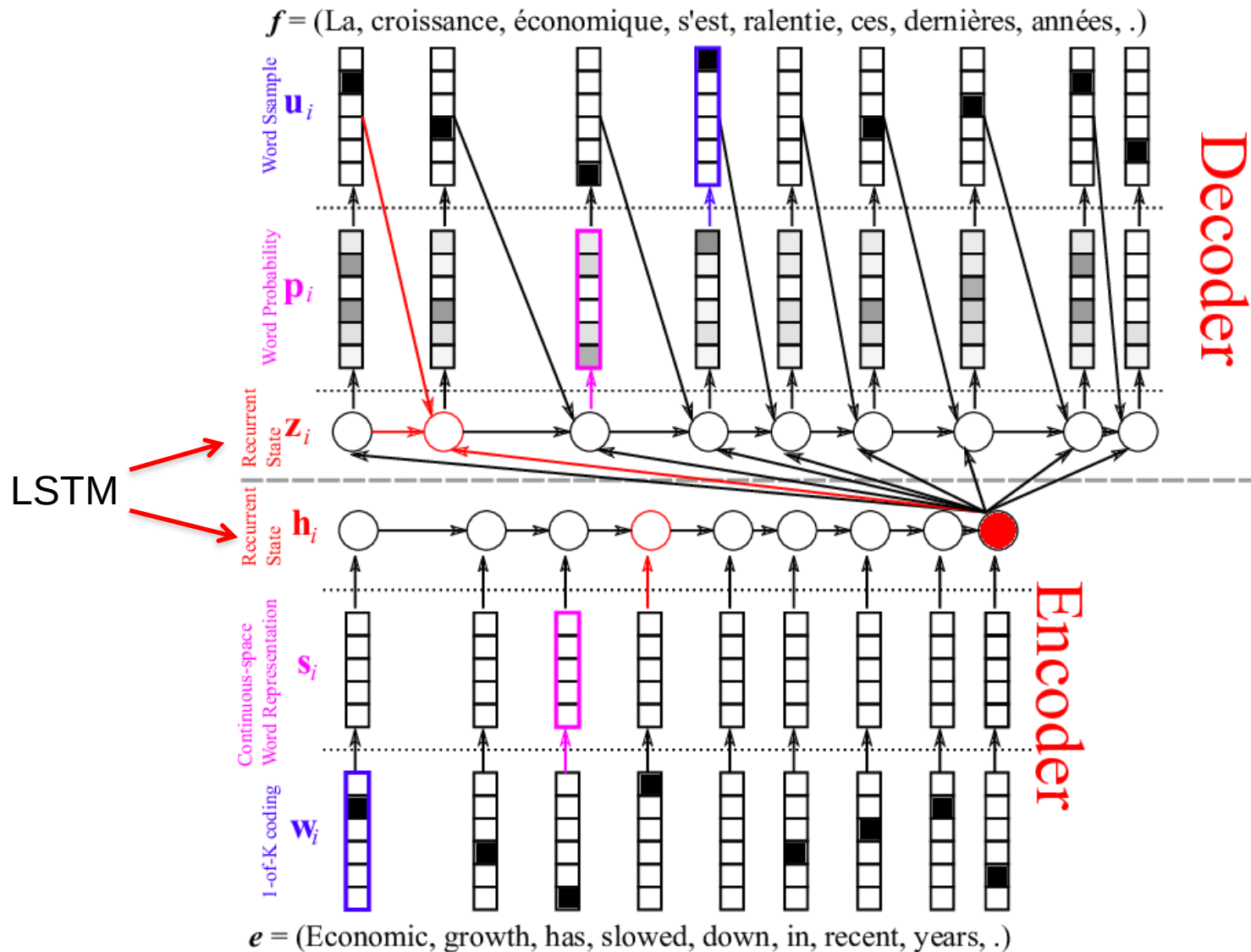
$$a^t = f(a^{t-1}, x^t) = \sigma(W^h a^{t-1} + W^i x^t + b^i)$$

t is time step

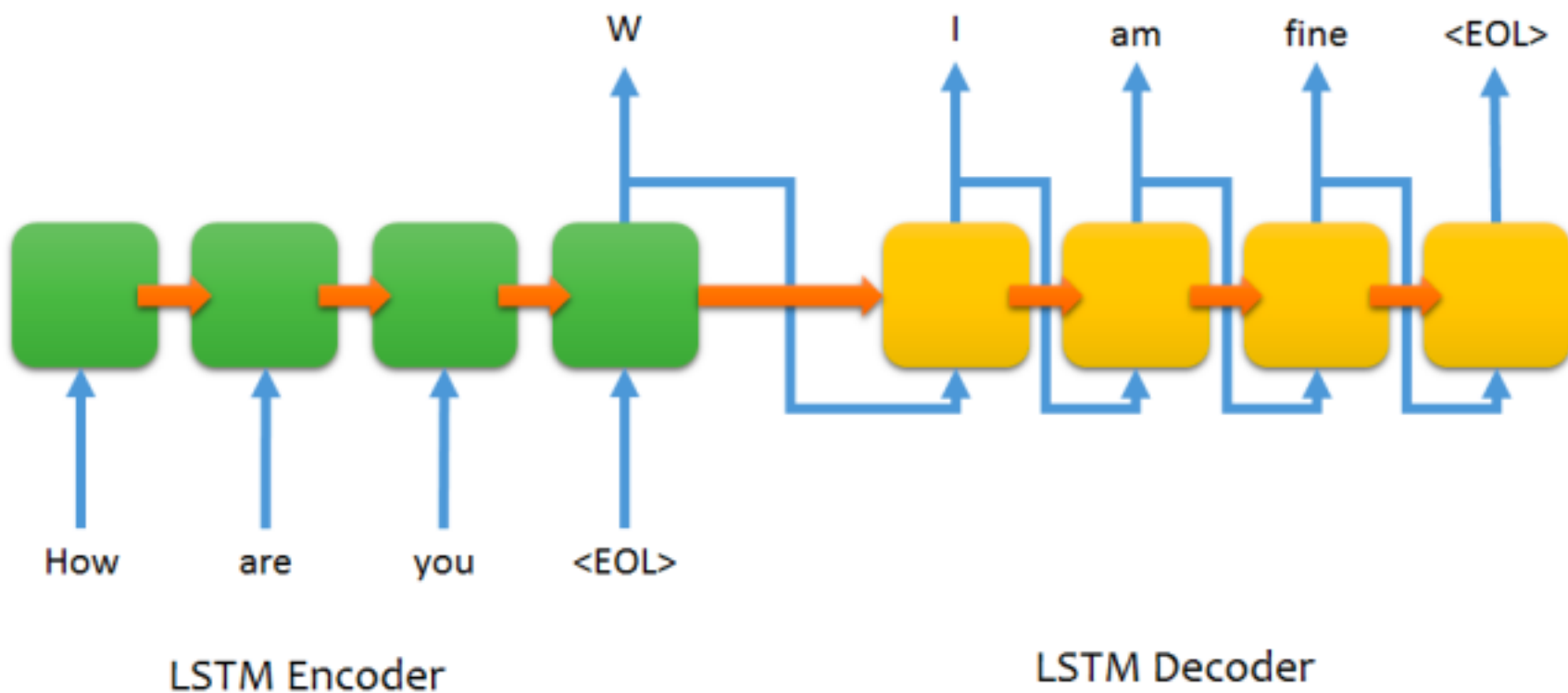
We will turn the recurrent network 90 degrees.

Applications of LSTM / RNN

Neural machine translation



Sequence to sequence chat model



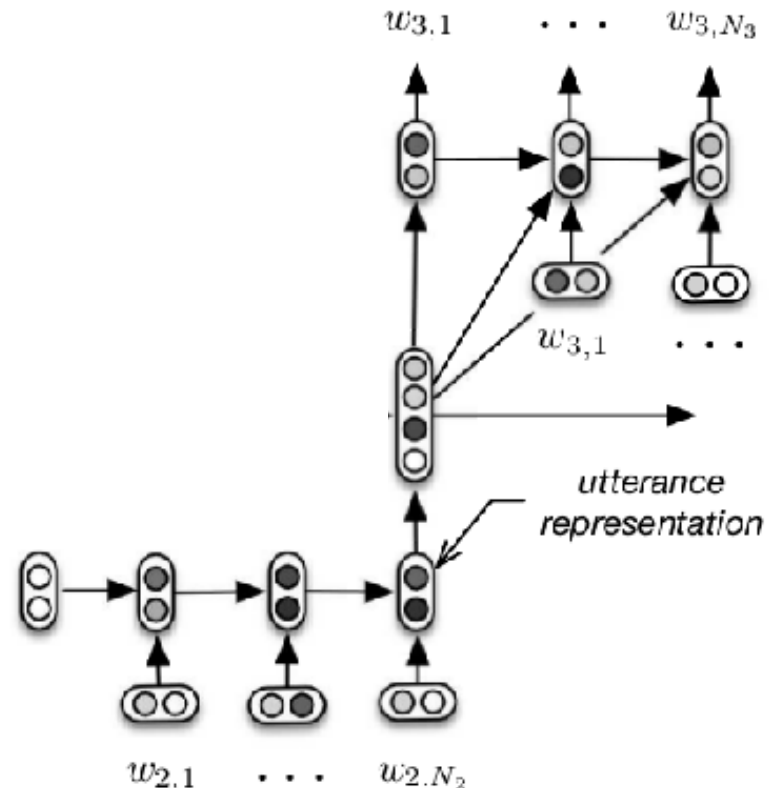
Chat with context

M: Hello

U: Hi

M: Hi

M: Hi 

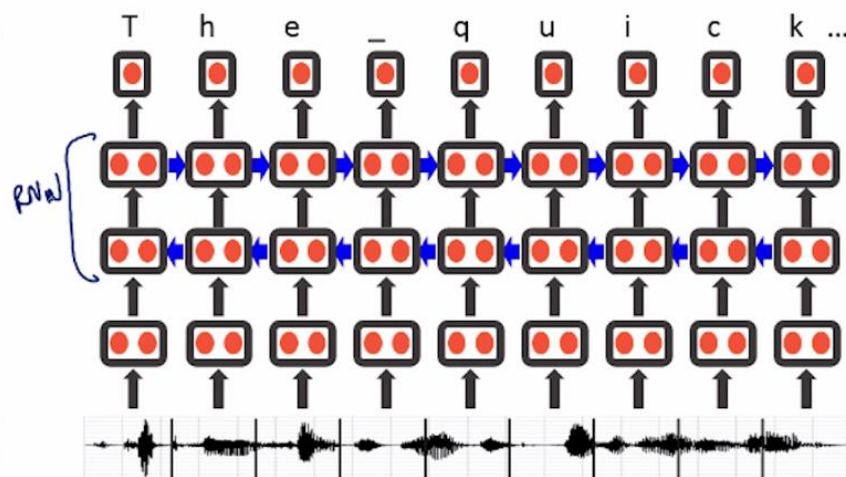
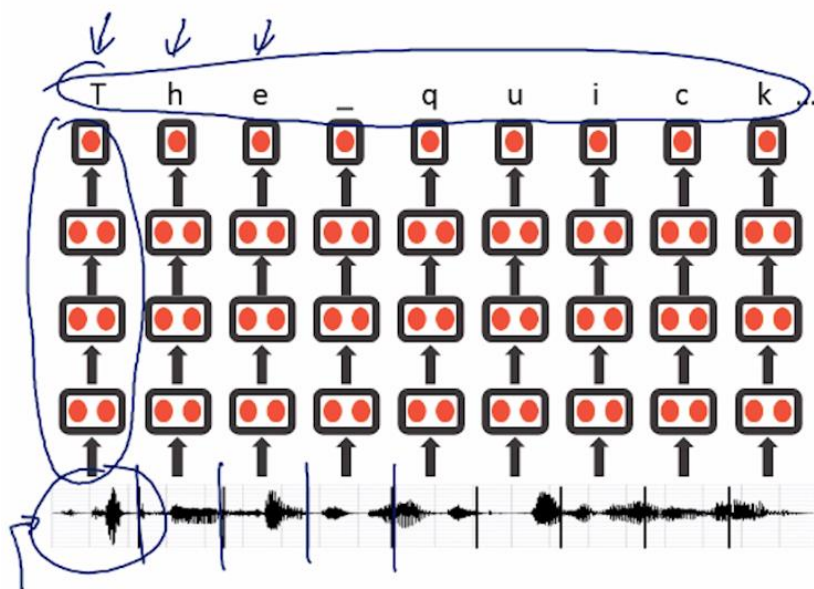


M: Hello

U: Hi

Baidu's speech recognition using RNN

Speech recognition example (Deep Speech)



Attention

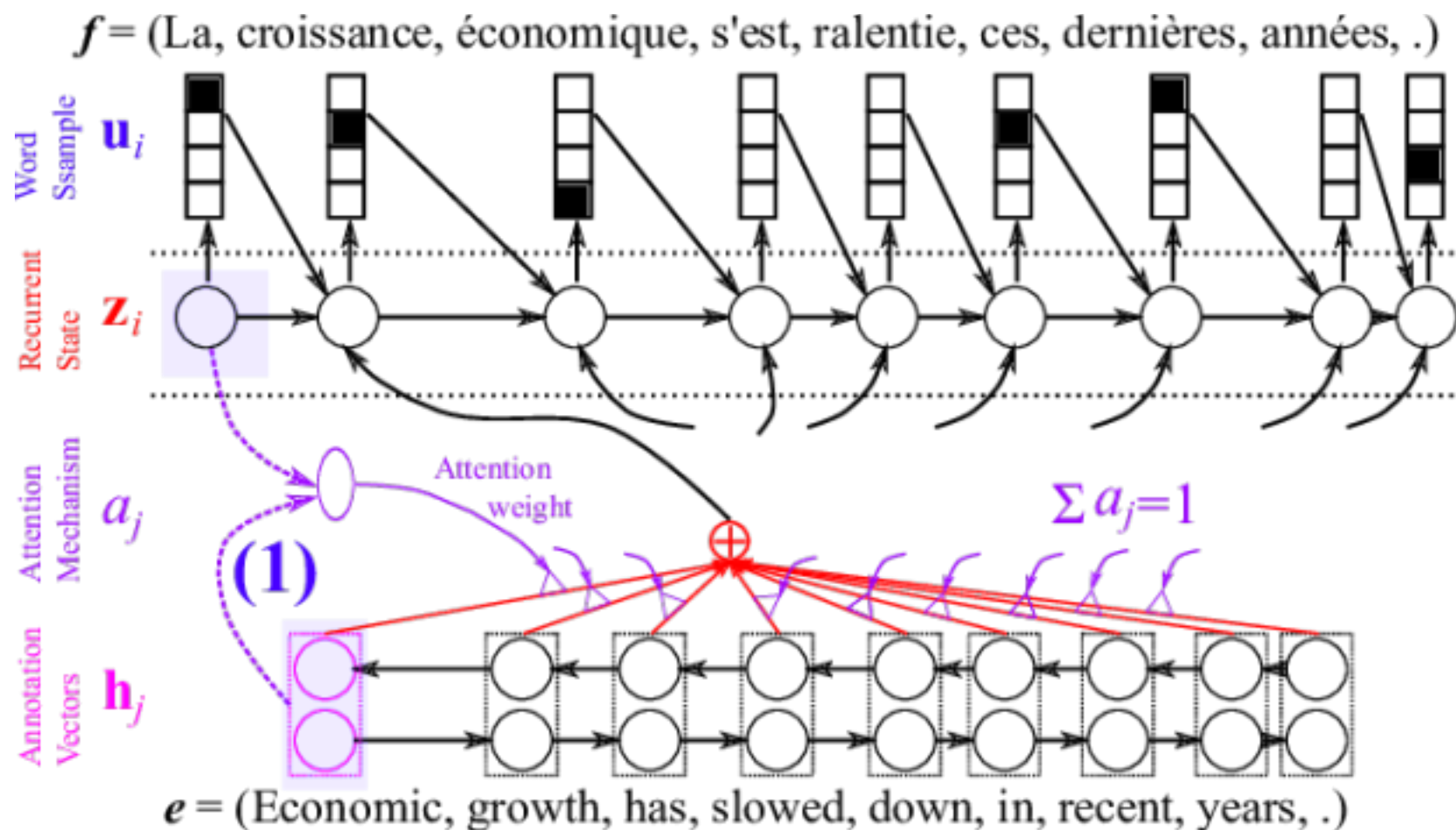


Image caption generation using attention

(From CY Lee lecture)

z^0 is initial parameter, it is also learned

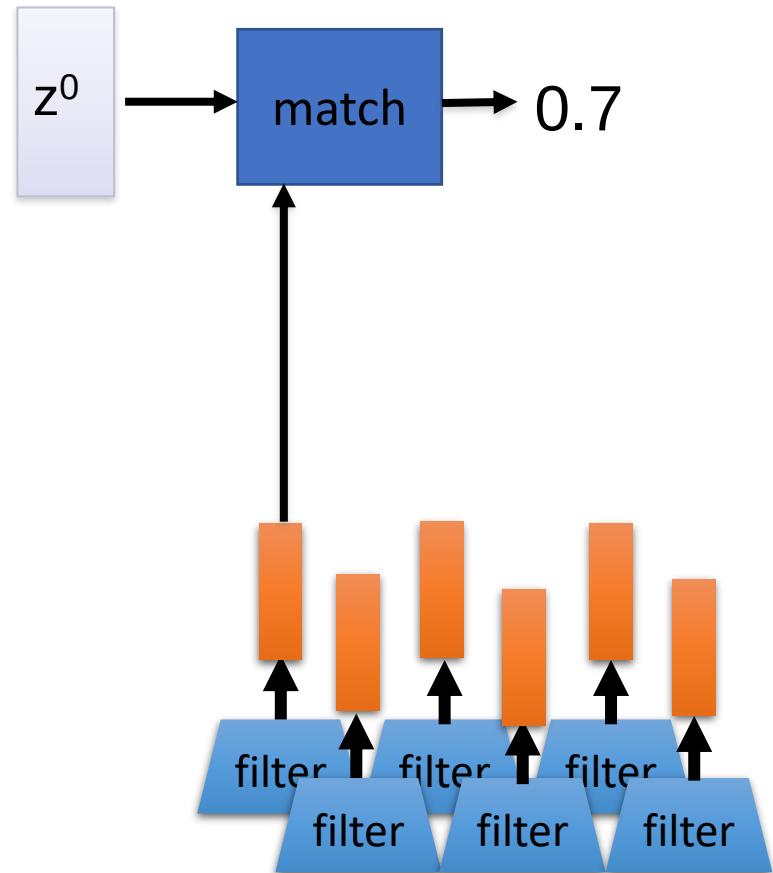
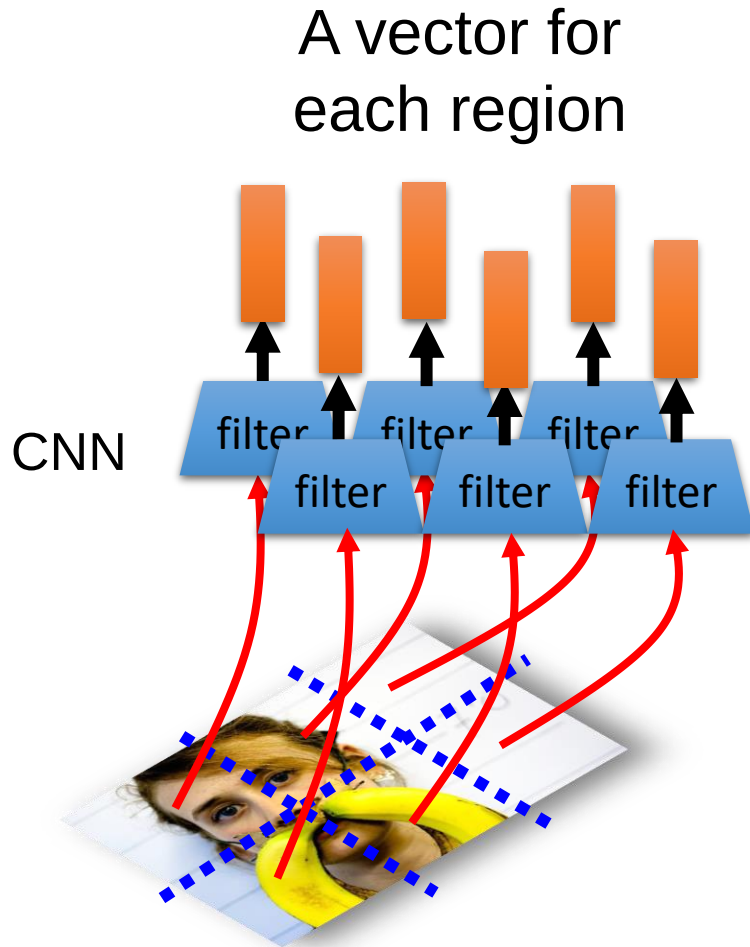


Image Caption Generation

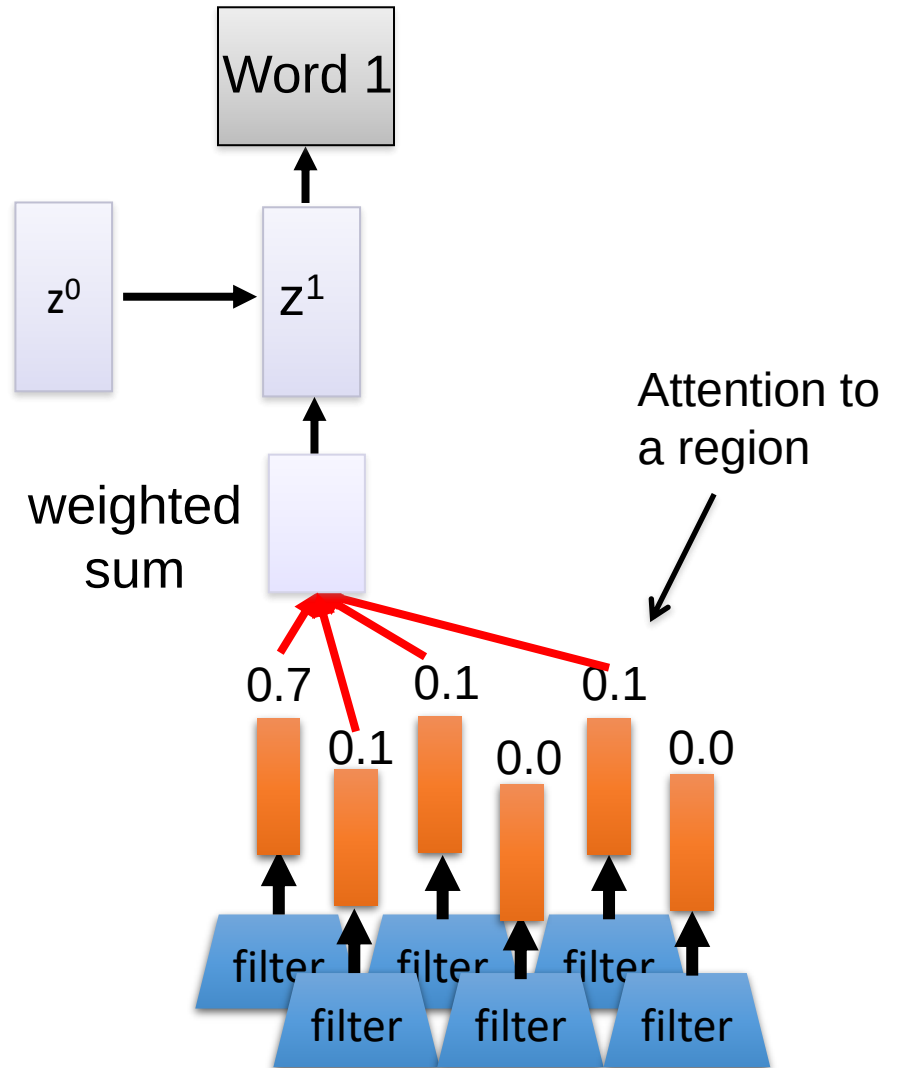
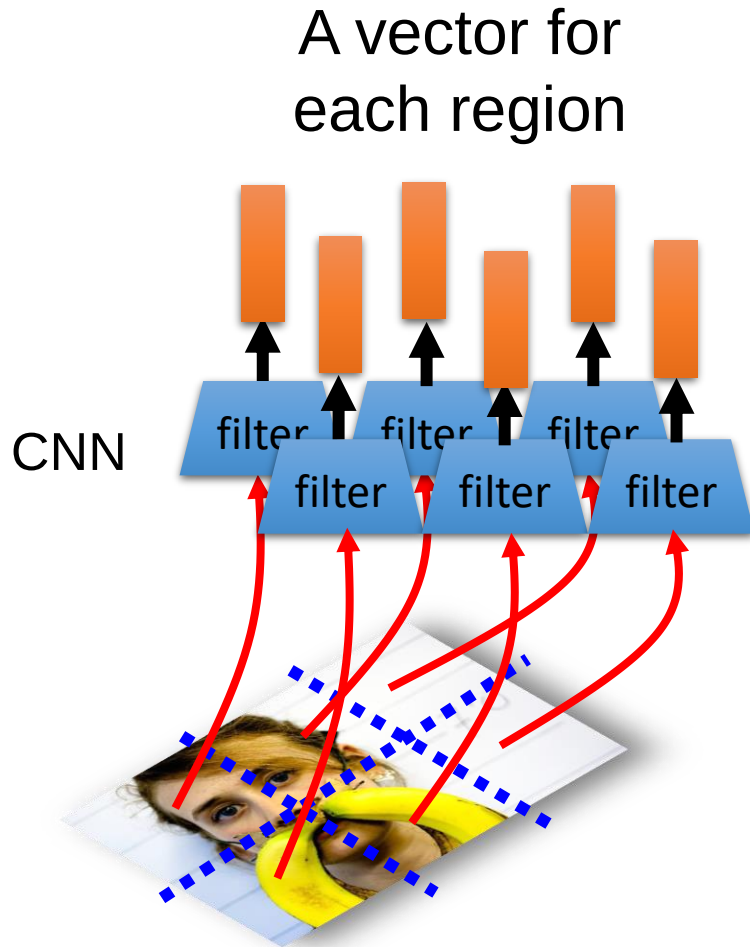


Image Caption Generation

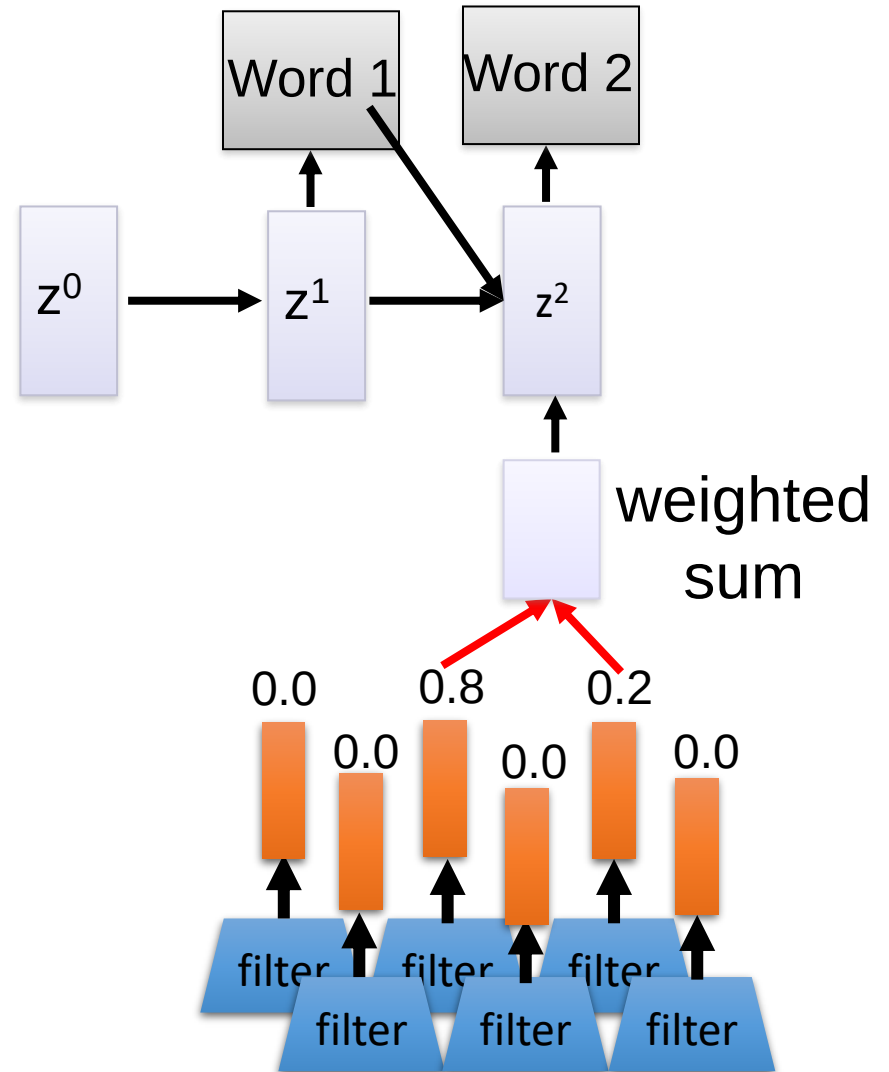
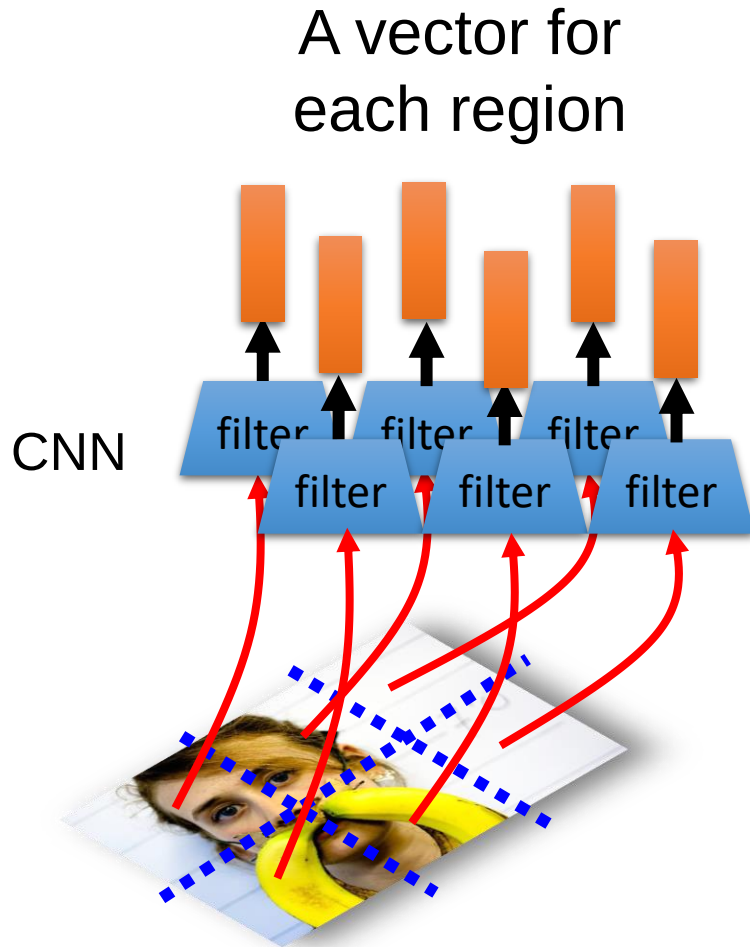
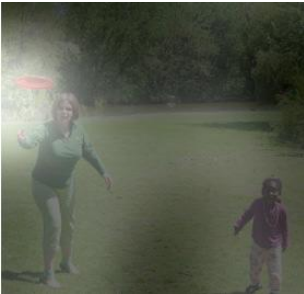


Image Caption Generation



A woman is throwing a frisbee in a park.



A dog is standing on a hardwood floor.



A stop sign is on a road with a mountain in the background.



A little girl sitting on a bed with a teddy bear.



A group of people sitting on a boat in the water.



A giraffe standing in a forest with trees in the background.



Kelvin Xu, Jimmy Ba, Ryan Kiros, Kyunghyun Cho, Aaron Courville, Ruslan Salakhutdinov, Richard Zemel, Yoshua Bengio, "Show, Attend and Tell: Neural Image Caption Generation with Visual Attention", ICML, 2015

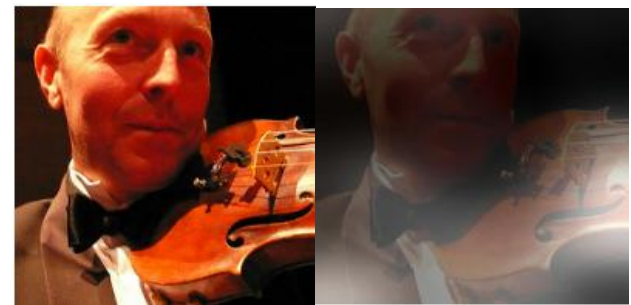
Image Caption Generation



A large white bird standing in a forest.



A woman holding a clock in her hand.



A man wearing a hat and a hat on a skateboard.



A person is standing on a beach with a surfboard.



A woman is sitting at a table with a large pizza.



A man is talking on his cell phone while another man watches.

Kelvin Xu, Jimmy Ba, Ryan Kiros, Kyunghyun Cho, Aaron Courville, Ruslan Salakhutdinov, Richard Zemel, Yoshua Bengio, "Show, Attend and Tell: Neural Image Caption Generation with Visual Attention", ICML, 2015



Ref: A man and a woman ride a motorcycle

A **man** and a **woman** are **talking** on the **road**



* Possible project?

Ref: A woman is frying food

Someone is **frying** a **fish** in a **pot**

Li Yao, Atousa Torabi, Kyunghyun Cho, Nicolas Ballas, Christopher Pal, Hugo Larochelle, Aaron Courville, "Describing Videos by Exploiting Temporal Structure", ICCV, 2015